



PRACTICAL AI IN HEALTHCARE

Healthy AI Implementation Roadmap for Hospitals

From Pilot to Production — A Step-by-Step
Playbook for Hospital Leaders and Analysts



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BOOK THREE

HEALTHY AI IMPLEMENTATION ROADMAP FOR HOSPITALS

*From Pilot to Production — A Step-by-Step Playbook for Hospital Leaders
and Analysts*

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Healthy AI Implementation Roadmap for Hospitals

Practical AI in Healthcare — Book 3

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Always keep qualified human professionals accountable for consequential healthcare decisions, including decisions influenced by AI.

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How to Use This Book

This is a working book, not a reference work. It is arranged in the order an implementation actually runs: decide how you will govern before you choose a project, choose the project before you look at vendors, prove the data before you sign anything, run a pilot that can fail honestly, and only then talk about scale.

Every chapter follows the same shape. It states the principle, describes how the failure actually happens inside a hospital, gives you the instrument — a matrix, a rubric, a gate, a checklist — and ends with the red flags that tell you the instrument is missing. The diagrams are not decoration. Each one is a page you can put on a screen in a steering committee meeting and point at.

You do not have to read it in order, but the instruments do build on one another. The selection matrix in Chapter 3 assumes the governance structure in Chapter 2 exists to use it. The pilot gates in Chapter 7 assume the data readiness score in Chapter 4 has been calculated. The scaling decision in Chapter 11 assumes the measurements in Chapter 10 were defined before the pilot began, not after it ended.

On numbers and rules, this book takes a deliberately careful line. Thresholds, weightings and scoring bands are stated plainly because they are conventions — sensible defaults you should adopt, argue with, and adapt to your own organisation. Regulatory requirements are described structurally, in terms of what has to exist and why, and the text sends you to the primary source rather than restating a date, a penalty figure or a certification requirement that may have moved. Nothing here has been independently verified against a live source for this edition, and the book should not be read as though it had been.

Where something depends on your jurisdiction, your regulator or your existing contracts, the text says so instead of inventing a universal answer. That is not evasion. It is the honest shape of the problem.

Introduction

Why Most AI Projects Never Reach Production

Ask around any hospital that has tried artificial intelligence and you will usually hear the same shape of story: a pilot that went rather well, and a deployment that never happened.

You will also find widely circulated statistics about the share of AI projects that die between pilot and production. I am not going to quote one. I cannot verify it for you, the figure changes with whoever is counting and what they chose to count, and — more importantly — the number is not the useful part. What is useful is that the failures are boringly consistent. They fail in the same places, for the same reasons, in the same order.

And almost none of those reasons are technical.

The algorithms work well enough. The computing capacity is available and affordable. The data exists somewhere in your estate. What goes wrong is organisational. Hospitals start AI projects without governance, so nobody can make a decision when a decision is needed. They select projects from vendor demonstrations rather than clinical need, so the tool solves a problem nobody was losing sleep over. They skip data readiness, so the model that worked on a hand-cleaned extract collapses on live data. They ignore change management, so the tool is bought, installed, and quietly not used. And because nobody agreed in advance what success would look like, there is no way to prove the thing helped — which makes it easy to abandon and move on to the next initiative.

I have watched this pattern repeat, not as an AI researcher and not as a vendor, but as a business analyst sitting in the rooms where these decisions get made. The difference between the hospitals that get AI into production and the ones that do not is almost never the technology they picked. It is whether they had a roadmap.

This book is that roadmap. It is a step-by-step playbook for hospital leaders, clinical leads and business analysts who want to implement AI as a sustainable, measurable, compliant operational capability rather than as a science experiment with a launch event.

What you will get

- A governance structure that makes AI decisions accountable to somebody specific, with the authority to stop a project as well as start one.
- A weighted selection matrix that lets you compare unlike projects honestly — and, just as importantly, lets you say no in public without it being personal.
- A data readiness assessment you can complete before you spend money, with thresholds that tell you plainly when not to proceed.
- A make-versus-buy decision path, and a procurement process that narrows the field on evidence rather than on sales pressure.
- A pilot design with three explicit gates, pre-agreed success criteria and a written way to fail.
- An integration standard that puts AI output where clinicians already look, and a change-management approach that treats resistance as information rather than obstruction.
- A measurement framework built on outcomes, with leading indicators you can watch weekly instead of waiting a year to be disappointed.
- A staged scaling approach, four compliance and ethics gates, and pause and rollback protocols written before you need them.
- A twelve-month roadmap that ends with one proven capability rather than twelve abandoned pilots.

What you will not get

No vendor recommendations. No technology comparisons. No claim that any particular product is safe, compliant or effective in your setting — that is a judgement only your own assessment can make. And no promises of easy wins, because the hospitals that believed in easy wins are the ones supplying the failure stories in Chapter 1.

What you get instead is a framework for thinking, a process for acting, and instruments you can put to work this month.

One idea to carry through the whole book

AI is not a project. It is a capability. Projects have end dates; capabilities have owners, budgets, monitoring and a maintenance burden that does not stop when the consultant leaves. Almost every failure described in this book comes from an organisation treating the second thing as though it were the first.

Capabilities require roadmaps. Wishlists are not roadmaps.



Figure I.1 — The shape of the book. Five parts, four decision gates. Each gate is a point at which the steering committee can stop the work, and stopping early is always cheaper than stopping late.

How the parts fit together

Part I is everything that has to be true before a project starts: why programmes fail, the governance structure that prevents it, and the matrix that decides which idea gets your attention. Part II is the foundation — data readiness, the build-or-buy decision, and the procurement process. Part III is the pilot itself: scope, gates, workflow integration and the people who have to use the thing. Part IV is scale — measuring return honestly, expanding in phases, and passing the compliance and ethics gates. Part V is what most books leave out: how to stop, how to pause, how to roll back with dignity, and a month-by-month roadmap for the first year.

Read Part I even if you are already mid-project. Most rescues start by going back and doing Chapter 2 properly.

PART I

Before You Start

Chapter 1 — Why AI Programmes Fail

The pattern that repeats

Failed AI programmes in hospitals are remarkably similar to one another. They are not each unhappy in their own way. They move through the same six stages, and once you have seen the sequence a few times you can predict the ending from the beginning.

1. **Excitement.** Somebody in leadership reads an article, attends a conference or sits through a vendor pitch. The conclusion is “we need to do AI”, which is a statement about identity rather than about a problem.
2. **Selection.** A project is chosen — usually the one with the most convincing demonstration, rarely the one attached to the most painful problem.
3. **Pilot.** A small team runs a limited trial. It produces promising results on a small, clean, carefully prepared dataset.
4. **Expansion.** Leadership, encouraged, asks to scale. But the pilot was designed to succeed, not to scale, and those are different designs.
5. **Friction.** Clinicians resist because nobody asked them. Data quality problems surface. Integration turns out to be the whole job. Compliance asks a question nobody has an answer to.
6. **Abandonment.** The project is not cancelled; it is quietly shelved. The team moves to the next initiative, and the organisation learns nothing because nobody writes down what happened.

The cycle then repeats with the next initiative, often within the same eighteen months, often with the same people, and usually without anyone naming it as a repeat.

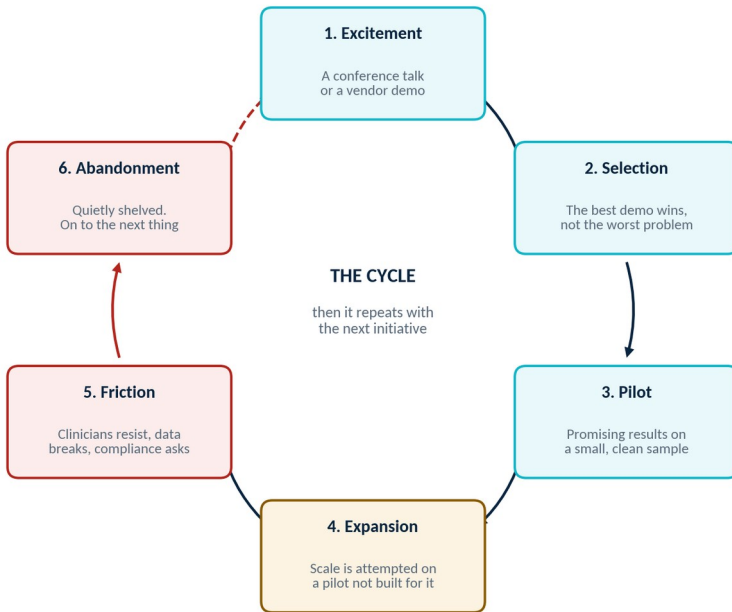
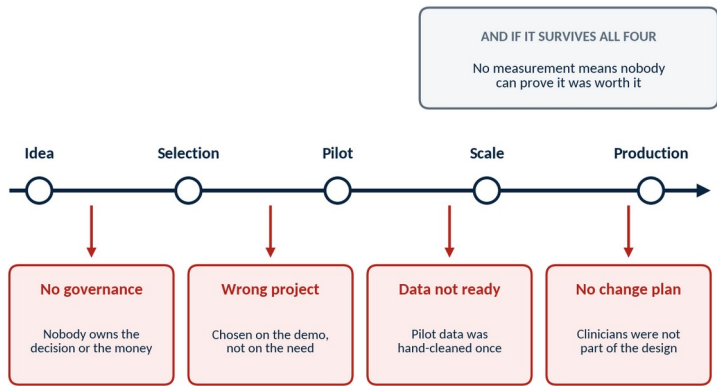


Figure 1.1 — The six-stage failure cycle. Nothing here is anyone's fault individually, which is exactly why it survives so well: the cycle has no owner and therefore no opponent.

The root of it is a category error. Hospitals treat AI as a technology project, so they staff it, budget it and govern it like one. It is an organisational change that happens to involve technology, and organisational change has entirely different failure modes.

Where projects actually die

It is worth being precise about location, because the remedy differs depending on where the project dies. Four causes kill projects outright at four identifiable points, and a fifth one does something more insidious: it lets the project survive while making it impossible to prove the project was worth anything.



Four causes kill the project. The fifth stops you proving it ever worked.

Figure 1.2 — Four causes kill the project outright. The fifth lets it live and then prevents anyone from proving it ever worked, which is how successful pilots still end up cancelled.

Root cause 1 — No governance

There is no committee, no named owner and no decision-making process. The project runs on the enthusiasm of one or two people. That works beautifully right up until the first genuine conflict, at which point it becomes clear that nobody has the authority to resolve anything. Budget questions escalate to whoever happens to be available. Departmental disagreements go unresolved because there is no forum in which to have them. And when the champion changes role or leaves the organisation, the project leaves with them.

The symptom to watch for is a project that can only move forward when one specific person is in the room.

Root cause 2 — The wrong project

The project was selected on vendor capability rather than clinical need. The demonstration was impressive; the sales team was responsive; leadership was excited. But the tool addresses a problem that clinicians were not complaining about, and the organisation now has to manufacture demand for its own solution.

This failure is unusually expensive because it looks like success for a long time. The implementation goes fine. The technology works as advertised. Usage is simply low, and it stays low, and eighteen months later somebody notices the licence renewal.

Root cause 3 — The data was not ready

The pilot worked because somebody prepared the data by hand. They pulled an extract, corrected the obvious errors, filled in the missing fields from a second system, and produced a dataset far cleaner than anything the organisation routinely holds. The model learned from that. Then it met production data — inconsistent, incomplete, spread across several systems, recorded differently by each department — and performed nothing like it did in the trial.

Nobody lied. The pilot result was real. It was just a result about a dataset that does not exist in daily operation.

Root cause 4 — No change management

Clinicians were not involved from the start, so they do not understand what the tool does, do not trust its output, and have no reason to change a workflow that currently works well enough. The tool sits in a corner of the interface being ignored. Adoption figures are reported as a training problem, which they are not — they are a design problem, and the design decision that caused them was made months earlier when nobody asked a clinician what would make this worth their time.

Root cause 5 — No measurement

The pilot produced “promising results”. Nobody defined in advance what success would look like. There is no baseline of the current state, so there is nothing to compare against. The reporting describes activity — predictions generated, users trained, sessions logged — rather than outcomes.

This is the cause that turns a genuinely useful tool into a cancelled line item. When budgets tighten, anything that cannot demonstrate its value is indistinguishable from anything that has none.

The prevention framework

Each root cause has a specific, unglamorous countermeasure. None of them are difficult. All of them have to be in place before the project starts, because every one of them is cheap in advance and expensive to retrofit.

Table 1.1 — The five root causes and their countermeasures. Read the third column carefully: the timing is the whole point.

Root cause	Prevention mechanism	When it must exist
No governance	An AI steering committee with a named executive sponsor and real authority to stop work	Before the first project is discussed

Root cause	Prevention mechanism	When it must exist
Wrong project selected	A weighted selection matrix scoring clinical need, feasibility and return	Before any vendor conversation
Data not ready	A data readiness assessment with a published threshold below which nothing proceeds	Before the pilot is designed
No change management	A named clinical champion and a workflow integration plan agreed with users	Before the tool is configured
No measurement	Pre-defined success criteria and a measured baseline of the current state	Before the pilot starts collecting data

Why this approach is called “healthy”

The method in this book is called a healthy implementation because it optimises for sustainability rather than speed. That is an unfashionable choice and it is worth being explicit about the trade-off: this is not the fastest way to get an AI tool into your hospital. It is the way that leaves you with something still working, still funded and still trusted two years later.

A healthy implementation has clear governance and accountability. It selects projects on evidence rather than excitement. It validates data readiness before committing resources. It involves clinicians from the first week rather than the last. It measures outcomes rather than activity. It scales gradually, proving value at each stage before widening the scope. And it has failure and rollback protocols written down before they are needed, so that stopping is a decision rather than a defeat.

RED FLAGS

- The business case begins with the technology rather than the problem.
- Nobody can name the person who is allowed to cancel the project.
- The pilot dataset was prepared by a person rather than a pipeline.
- No clinician was in the room when the workflow was designed.
- Success is described in adjectives — “promising”, “encouraging” — rather than in numbers against a baseline.
- The organisation has done this before and cannot say why the last attempt stopped.

CHAPTER 1 TAKEAWAY

AI programmes fail organisationally, not technically, and they fail in five predictable ways. Every countermeasure in this book is cheap before the project starts and expensive afterwards. If you fix only one thing, fix governance — it is the cause that prevents you from fixing the others.

Chapter 2 — Building the AI Governance Structure

Why governance matters more than it sounds like it should

Governance is the least interesting word in this book and the most load-bearing. Without it, AI projects become hobbies. They run on individual enthusiasm rather than organisational commitment, they succeed or fail on the persistence of one person, and they die when that person moves on.

Governance is what converts a series of interesting experiments into a programme. Specifically, it supplies four things that enthusiasm cannot. It supplies decision-making authority: somebody who can approve a project, allocate a budget and resolve a dispute between two departments who both want to go first. It supplies accountability: a named person answerable for the outcome and obliged to report on it. It supplies standards: the criteria a project must meet and the process it must follow, applied to everybody equally. And it supplies oversight: somebody watching progress closely enough to catch a problem while it is still small.

None of that requires a large bureaucracy. It requires one committee that actually meets and actually decides.

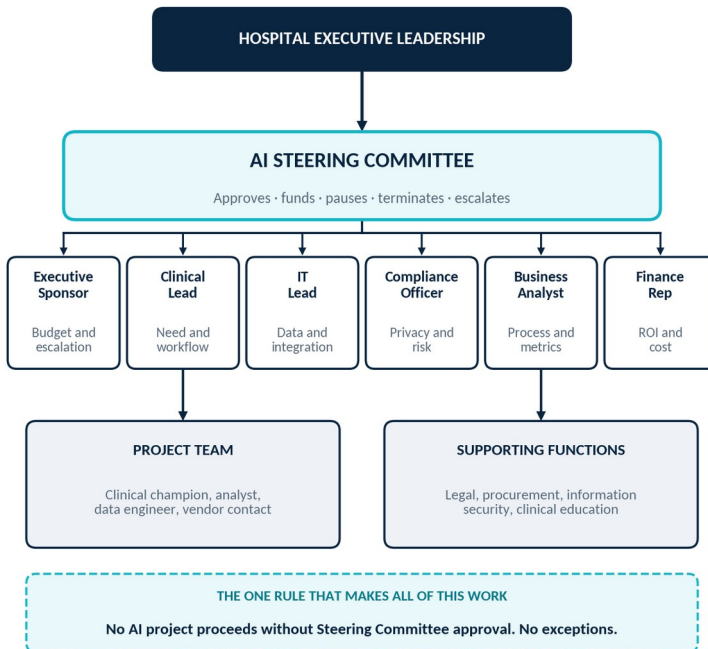


Figure 2.1 — The governance structure. The committee sits between executive leadership and the project team, and it holds the only authority that matters: the authority to stop.

The AI steering committee

The core governance body is the AI steering committee. It is not an advisory group, a working party or a suggestion box. It is a decision-making body with budget authority, and if it does not have budget authority it will be routed around within two months.

Six roles need to be represented. You can combine them in a smaller organisation, but you cannot leave a perspective out — every missing seat becomes a category of problem that surfaces late.

Table 2.1 — Composition of the AI steering committee. Combine roles if you must; do not omit a perspective.

Role	What they are there for
Executive sponsor (C-suite)	Budget authority, organisational alignment, and the escalation route when two departments disagree
Clinical lead (physician or nurse leader)	Validating that the need is real, judging workflow fit, and carrying credibility with clinical staff
IT lead (CIO or IT director)	Infrastructure, security, integration effort and realistic data access timelines
Compliance officer	Regulatory obligations, privacy assessments, risk appetite and the questions that must be answered before deployment
Business analyst	Requirements, process mapping, metrics, vendor coordination and keeping the paperwork honest
Finance representative	Budget approval, total cost of ownership, and holding the return projections to account after the fact

Cadence and authority

Monthly is the minimum workable cadence, with a longer strategic review each quarter. Less often than monthly and the committee becomes a reporting ceremony, because decisions will already have been made informally by the time it convenes.

The committee’s authority should be written down and should include all of the following: approving and rejecting new AI projects; allocating budget and staff; resolving conflicts that cross departmental boundaries; pausing or terminating projects that are not performing; and approving every scaling decision. The power to terminate is the one most often left out and the one that makes the rest credible.

The governance charter

Every AI programme needs a short charter — two or three pages, not twenty — that fixes the things people otherwise argue about later. At minimum it should record the committee’s purpose and scope, its

membership by role rather than by name, its meeting cadence and quorum, the decisions reserved to it, the escalation path above it, the criteria a project must meet to be considered, the reporting it expects, and the review date at which the charter itself is reconsidered.

Write it before your first project, not during your first crisis. A charter drafted in the middle of a dispute is a negotiating position, and everyone in the room will read it as one.

The RACI matrix

For each project, agree who is responsible for doing the work, who is accountable for the outcome, who must be consulted before a decision, and who must be informed after it. The distinction between responsible and accountable is the one that saves you: they are different people, and the second one cannot be a committee if you want anything to happen.

Table 2.2 — A worked RACI for a hospital AI project. Adapt the names to your structure, but keep exactly one accountable party per row.

Activity	Responsible	Accountable	Consulted	Informed
Project selection	Business analyst	Steering committee	Clinical lead, IT lead	Department heads
Data readiness	Data team	IT lead	Compliance officer	Clinical lead
Vendor selection	Business analyst	Steering committee	IT lead, compliance	Finance
Pilot execution	Project team	Clinical lead	IT lead	Steering committee
Change management	Clinical champion	Clinical lead	Business analyst	All affected staff
Return measurement	Business analyst	Finance representative	Clinical lead	Steering committee
Scaling decision	Steering committee	Executive sponsor	All roles	Department heads

The rule that makes all of this work

One sentence does more than the rest of the charter combined: no AI project proceeds without steering committee approval, and there are no exceptions.

This is what closes off shadow AI — departments quietly running their own tools, often with real patient data, outside any oversight. Shadow AI is rarely malicious. It happens because the official route is slow and the unofficial route is a browser tab. Two things suppress it: an approval

process fast enough to be worth using, and a rule everybody knows. If your approval process takes three months, you have not eliminated shadow AI; you have merely stopped hearing about it.

Governance readiness checklist

- ☐ The steering committee exists and has met at least twice
- ☐ An executive sponsor is named and attends in person
- ☐ All six perspectives are represented, even if roles are combined
- ☐ The charter is written, circulated and dated
- ☐ The committee's authority explicitly includes pausing and terminating
- ☐ A RACI exists for each active project with one accountable name per row
- ☐ The approval route is documented and takes weeks, not months
- ☐ Staff know the rule about unapproved AI tools without having to look it up

RED FLAGS

- The committee meets but never rejects anything.
- The executive sponsor sends a delegate every time.
- Projects reach procurement before the committee has seen them.
- Nobody can produce the charter.
- Compliance is consulted after the contract is drafted.
- Two departments are running similar AI tools and neither knew.

CHAPTER 2 TAKEAWAY

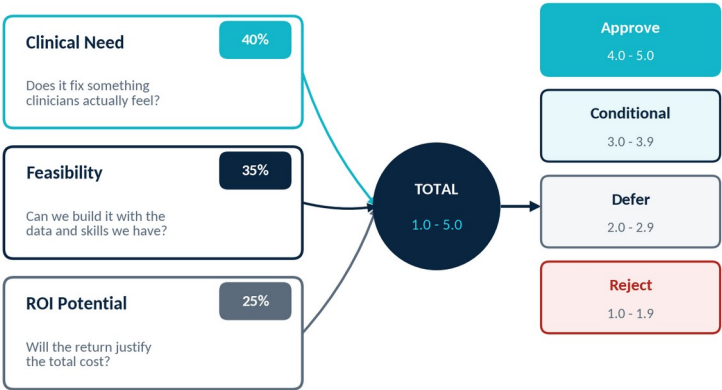
One committee, six perspectives, monthly, with the authority to stop work and a named executive sponsor who turns up. Write a short charter before the first project. Make the approval route fast enough that nobody has a reason to go around it.

Chapter 3 — The Project Selection Matrix

Picking wins, not hype

Hospital leadership is bombarded with AI opportunities. Every vendor claims their product will transform care. Every conference programme features a success story. Every second article promises a revolution in something.

The predictable result is that projects get chosen on excitement rather than evidence, and the shiniest demonstration beats the most painful problem. The remedy is not more scepticism, which simply makes decisions slower and more arbitrary. The remedy is a scoring instrument that everybody agreed to before anyone had a favourite.



$$\text{Total} = (\text{Need} \times 0.40) + (\text{Feasibility} \times 0.35) + (\text{ROI} \times 0.25)$$

For every project you approve, expect to defer or reject at least two.

Figure 3.1 — The weighted selection matrix. Three dimensions, fixed weights, one number, four decision bands. Agree the weights before you score anything.

The three dimensions

Candidate projects are scored on clinical need, feasibility and return potential, each from one to five, and combined with fixed weights. The weights below — forty, thirty-five and twenty-five per cent — encode a deliberate position: that a real clinical need is worth more than an attractive business case, and that both are worthless if the project cannot actually be built with the data and skills you have.

You may reasonably choose different weights. What you may not do is change them after you see the scores.

Dimension 1 — Clinical need (weight 40 per cent)

This asks whether clinicians would notice if the problem went away. Score it with a clinician in the room, not for them.

Table 3.1 — Clinical need scoring anchors.

Score	What it means
5	Addresses a critical patient safety or care quality issue
4	Addresses a significant operational inefficiency that clinical staff complain about unprompted
3	Addresses a moderate clinical or operational need
2	Addresses a minor need or a convenience
1	No clear clinical or operational need; the interest is in the technology itself

Dimension 2 — Feasibility (weight 35 per cent)

This asks whether you could build or deploy it with what you have now. Feasibility is where optimism does the most damage, so score it with the IT lead and the data team present and let them be pessimistic.

Table 3.2 — Feasibility scoring anchors.

Score	What it means
5	Data is available, clean and accessible; the technology is proven; integration is straightforward
4	Data is mostly available; the technology is proven; integration needs moderate effort
3	Data needs significant preparation; the technology exists; integration is complex
2	Data is limited or unavailable; the technology is emerging; integration is very complex
1	The data does not exist; the technology is experimental; integration is not feasible today

Dimension 3 — Return potential (weight 25 per cent)

This asks whether the benefit justifies the total cost of ownership — not the licence fee, the total. The bands below are a convention for comparing candidates against each other, not a promise about what any project will deliver.

Table 3.3 — Return potential scoring anchors. Score the total cost of ownership, not the licence price.

Score	What it means
5	Strong return expected within twelve months; clear cost saving or revenue impact
4	Solid return expected within twelve months; significant cost or revenue impact
3	Moderate return expected within eighteen months
2	Return below cost within twenty-four months; minimal financial impact
1	No identifiable return; purely exploratory

Scoring and thresholds

The total is a simple weighted sum:

THE FORMULA

Total = (Clinical need × 0.40) + (Feasibility × 0.35) + (Return × 0.25). The result lands between 1.0 and 5.0, and the band it falls into determines the decision — before anyone argues about it.

Table 3.4 — Decision bands. Publish these before the first scoring session so that the threshold is not negotiated against a favourite.

Total	Decision	What it actually means
4.0 – 5.0	Approve	Proceed to data readiness assessment and pilot design
3.0 – 3.9	Conditional	A specific gap must be closed and re-scored before the project moves
2.0 – 2.9	Defer	Revisit when the data, the market or your capability changes
1.0 – 1.9	Reject	Not suitable for AI in your organisation; say so plainly

A worked example

Five candidates, scored in a single session by the steering committee with the clinical lead and IT lead present. The interesting rows are the last three, where a strong score on one dimension is correctly overwhelmed by a weak score on another.

Table 3.5 — Worked scoring. The chatbot is highly feasible and still conditional, because feasibility is not a reason to do something.

Candidate project	Need	Feas.	Return	Total	Decision
Imaging support for pneumonia detection	5	4	4	4.4	Approve
Sepsis early warning	5	3	4	4.0	Approve

Candidate project	Need	Feas.	Return	Total	Decision
Chatbot for patient frequently asked questions	2	5	3	3.2	Conditional
Genomic analysis platform	4	1	2	2.5	Defer
Ambient medical transcription	3	3	2	2.7	Defer

Notice what the matrix does to the chatbot. It is the easiest project on the list — the data exists, the technology is mature, integration is trivial. It scores 3.2 because ease is not a clinical argument. Without the weighting, it would have been first in the queue simply because it was available, and the sepsis project, which is harder and matters more, would have waited another year.

The discipline of saying no

The matrix is at least as much about rejection as approval. Most hospitals have more AI opportunities than they have data engineers, clinical champions and change capacity — and it is that last constraint, not money, that usually binds.

A workable rule of thumb: for every project you approve, expect to defer or reject at least two. That ratio is not pessimism. It is what focus looks like when it is written down.

Saying no is also considerably easier when it is the instrument saying it rather than a person. A deferred project with a score, a recorded reason and a named condition for reconsideration is a professional outcome. A deferred project with no explanation is a grievance, and grievances come back as shadow AI.

RED FLAGS

- The weights were adjusted after the scores were known.
- Every candidate scores above four.
- Feasibility was scored without the data team present.
- A project entered the pipeline without a score because it was “already agreed”.
- Nothing has been rejected in twelve months.
- Deferred projects have no recorded condition for reconsideration.

CHAPTER 3 TAKEAWAY

Score every candidate on clinical need, feasibility and return, with the weights fixed in advance and the decision bands published. Expect to reject or defer two projects for every one you approve, and always record why — a documented no protects the programme far better than a quiet yes.

PART II

The Foundation

Chapter 4 — Data Readiness

The gap between pilot data and real data

The single most common reason an AI pilot fails to scale is data. Not model architecture, not integration, not budget — data. And the failure has a specific shape: the pilot succeeded on a dataset that a human being curated, and production runs on a dataset that nobody curates at all.

At scale, hospital data is messy in ways that are entirely normal and entirely fatal to a model trained on something tidier. Fields are completed differently by different departments. Free text carries information that the structured fields are missing. Records are duplicated across systems with slightly different spellings. Devices write timestamps in local time and applications write them in coordinated universal time. None of this is a scandal. It is what an operational estate looks like after twenty years of good-enough decisions.

So before committing money or clinical goodwill to any AI project, assess data readiness — deliberately, with a score, on paper. This is the cheapest gate in the entire book and the one most often skipped, because it produces answers people do not want during a period of enthusiasm.

SCORE EACH DIMENSION 1 - 5

Availability Is the data there, in a form we can reach?	1 2 3 4 5
Quality Is it accurate, complete and current enough?	1 2 3 4 5
Accessibility Can we get it out without heroics?	1 2 3 4 5
Governance Is anyone actually accountable for it?	1 2 3 4 5
Volume Is there enough of it to learn anything?	1 2 3 4 5

TOTAL 5 - 25

20 - 25 Ready → proceed	15 - 19 Prepare first	10 - 14 Major data work	5 - 9 Do not proceed
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Figure 4.1 — The data readiness assessment. Five dimensions, each scored one to five, giving a total between five and twenty-five. The threshold is published in advance so that it cannot be renegotiated afterwards.

How to run the assessment

Score each of the five dimensions from one to five against the anchors below. Do it as a group — data team, IT lead, business analyst and someone from the clinical area that generates the data — and record the reasoning next to each score, not just the number. Six months later, the reasoning is what tells you whether anything has actually changed.

Score what is true today. Not what is planned, not what the migration project will deliver next year, and not what the data would look like if everyone followed the documentation.

Dimension 1 — Availability

Table 4.1 — Data availability anchors.

Score	What it means
5	All required data exists in structured form in systems you can reach
4	Most required data is available; the remaining gaps are small and fillable
3	Significant data exists but must be extracted from several different systems
2	Limited data exists; substantial new collection would be required
1	The data does not exist, or exists somewhere you cannot get to

Dimension 2 — Quality

Table 4.2 — Data quality anchors. Measure the error rate on a sample rather than estimating it from memory.

Score	What it means
5	Accurate, complete, consistent and timely; error rate under one per cent
4	Mostly accurate and complete; error rate roughly one to five per cent
3	Moderate quality problems; error rate roughly five to fifteen per cent
2	Significant quality problems; error rate roughly fifteen to thirty per cent
1	Poor quality; error rate above thirty per cent, or critical fields routinely missing

Dimension 3 — Accessibility

Table 4.3 — Data accessibility anchors.

Score	What it means
5	Available through standard interfaces; no manual extraction needed
4	Available, but requires some manual extraction or transformation
3	Requires significant extraction, transformation and loading work

Score	What it means
2	Locked inside systems with limited export capability
1	Not accessible without modifying the source systems

Dimension 4 — Governance

Table 4.4 — Data governance anchors. A low score here is a warning about every future project, not just this one.

Score	What it means
5	Policies are documented and enforced; data stewards are named and active
4	Policies exist but enforcement is inconsistent
3	Some governance exists but coverage is partial
2	Minimal governance; arrangements are informal and undocumented
1	No data governance of any kind

Dimension 5 — Volume

Table 4.5 — Data volume anchors. Volume is judged against the rarity of what you are trying to detect, not against a fixed record count.

Score	What it means
5	Comfortably enough data for training and independent validation
4	Adequate for training; may need augmentation for rarer cases
3	Limited; may require transfer learning or synthetic augmentation
2	Very limited; model performance is likely to be unreliable
1	Insufficient for any credible training or validation

Reading the total

Add the five scores. The total falls between five and twenty-five, and the band it lands in is the decision.

Table 4.6 — Readiness thresholds. The bottom row is a genuine stop, not a strongly worded caution.

Total	Decision	What happens next
20 – 25	Ready	Proceed to pilot design
15 – 19	Prepare first	Close the specific gaps, re-score, then proceed
10 – 14	Major data work required	Treat data remediation as its own funded project; consider a different candidate meanwhile

Total	Decision	What happens next
5 – 9	Do not proceed	No AI project on this data will survive contact with production

The remediation plan

A score below twenty is not a rejection; it is a work order. Write a remediation plan that names, for each dimension scoring below four, the specific gap, the work required to close it, the owner, the target date, and the score you expect afterwards. Then re-score on that date with the same group.

Two warnings. First, remediation is real work with real cost, and it should be funded and tracked as such rather than absorbed quietly by a data team that already has a queue. Second, remediation that only fixes the extract is not remediation. If the correction happens downstream of the systems that generate the data, you have cleaned this dataset and changed nothing about the next one.

The thirty-day rule

Before starting any AI project, ask one blunt question: can we get the data we need, in the form we need it, within the next thirty days, without buying new software or hiring new people?

If the answer is no, the project is not ready for a pilot. That is not a reason to abandon it — it is a reason to sequence it correctly. Fix the data, then come back. Projects that ignore this question do not avoid the work; they simply discover it in month seven, with a vendor contract already signed and a clinical champion already committed.

Data readiness checklist

- ☐ All five dimensions scored, with reasoning recorded beside each score
- ☐ Scores reflect today’s data, not a planned future state
- ☐ The error rate was measured on a sample, not estimated
- ☐ The data team and a clinical data owner both took part in scoring
- ☐ The total is above the published threshold, or a remediation plan exists
- ☐ Remediation is funded, owned and dated
- ☐ Fixes are applied at source, not only to the extract
- ☐ The thirty-day question has been asked and honestly answered

RED FLAGS

- The pilot dataset was assembled by one person over a weekend.
- Nobody can say who owns the source data.
- The quality score is a guess.
- Remediation is described as “the data team will sort it out”.
- The assessment was completed after the vendor was chosen.
- A score below the threshold was overridden because of project momentum.

CHAPTER 4 TAKEAWAY

Score availability, quality, accessibility, governance and volume before you spend anything. Publish the threshold in advance and respect it. If you cannot get usable data within thirty days without new software or new staff, fix the data first — that sequence is the difference between a pilot that scales and a pilot that impresses.

Chapter 5 — Make, Buy or Skip

Three answers, not two

Every AI project faces the same decision, and it has three possible answers rather than the two people usually discuss: build it in-house, buy it from a vendor, or decline to do it at all.

The third answer is the one that gets left out of the meeting, which is unfortunate, because it is frequently the correct one. A make-or-buy discussion that cannot conclude “neither” is not a decision process; it is a procurement process wearing a disguise.

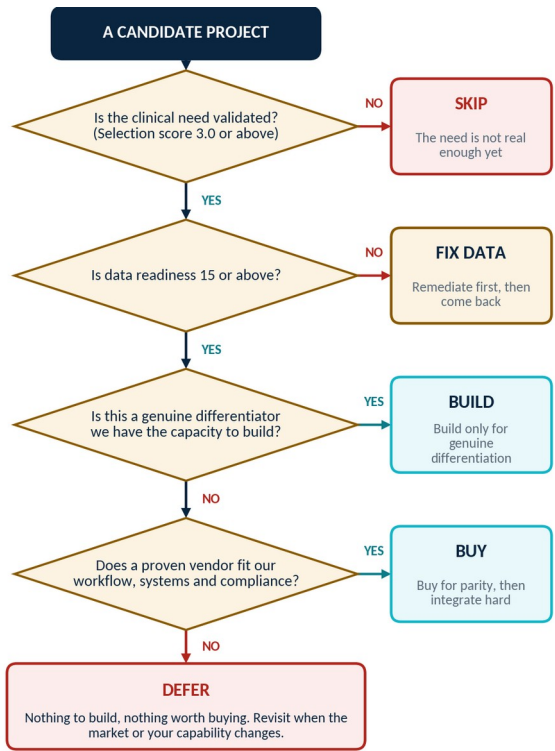


Figure 5.1 — The build, buy or skip decision path. Each question is a genuine stop, and the order matters: validate the need before you assess the data, and assess the data before you talk to anyone selling anything.

When to build

Build when artificial intelligence is, or is deliberately becoming, a core competency of your organisation. Build when your data is genuinely distinctive — an unusual patient population, a workflow no product

anticipates, a combination of sources nobody else holds. Build when you need customisation deeper than a vendor will accommodate, or when regulatory or contractual obligations demand that the model and the data stay under your direct control.

And build only when you honestly have the talent, the budget and the time — including the time after launch. A model is not a deliverable; it is a thing that degrades quietly and needs monitoring, retraining and a person who understands it available for years.

When to buy

Buy when a vendor has a proven solution for your specific use case and can show it working somewhere comparable. Buy when you need deployment inside six months. Buy when you lack internal AI engineering capability and are not trying to build it. Buy when the product integrates with the systems you already run rather than requiring you to change them. And buy only when the vendor can produce the compliance artefacts your compliance officer asks for — in writing, before signature, not as a promise for later.

When to skip

Skip when the clinical need is not validated and the selection score is low. Skip when data readiness is poor and cannot be remediated within a sensible horizon. Skip when no credible product exists and you cannot realistically build one. Skip when the return does not justify the total cost of ownership. And skip when the regulatory position is genuinely unresolved rather than merely unfamiliar — there is a difference, and your compliance officer can tell you which one you are looking at.

Skipping is not failure. Deferred projects with recorded reasons are an asset: they are the shortlist for next year, already half analysed.

Table 5.1 — Signals and what they point to. Where signals conflict, the skip signals outrank the others.

Signal	What it points to
The data is distinctive and cannot be replicated by a vendor	Build
The capability differentiates your organisation clinically	Build
You have engineering talent and a multi-year maintenance budget	Build
The use case is standard and several credible products exist	Buy
Deployment is needed within six months	Buy
There is no internal AI engineering capability	Buy
The clinical need scored below three on the selection matrix	Skip
Data readiness is below fifteen with no funded remediation	Skip

Signal	What it points to
The regulatory position is genuinely unresolved	Skip

Build for differentiation, buy for parity

The most durable rule in this chapter is a single sentence: build the capabilities that make your hospital different, buy the ones that merely keep you level.

Standard diagnostic support, routine administrative automation and common documentation tasks are parity capabilities. Being marginally better at them than the hospital across the city changes very little, and building them consumes engineering capacity you will want later. The capabilities worth building are the ones rooted in something only you have — your case mix, your referral patterns, your particular operational constraints.

Total cost of ownership

Neither option costs what the first estimate says. Compare them properly, over the same time horizon, with the same categories included on both sides.

Table 5.2 — Total cost of ownership on both sides. The bottom two rows are the ones routinely omitted, and they are frequently decisive.

Build — cost categories	Buy — cost categories
Development: staff, infrastructure, tooling	Licensing: annual, per-seat or per-use
Ongoing maintenance: monitoring, retraining, support	Integration: connecting to your systems and keeping it connected
Opportunity cost: what the team is not doing instead	Training: teaching staff to use it, and re-teaching after upgrades
Risk cost: the expected cost of the project not working	Vendor risk: price rises, acquisition, discontinuation, exit
Institutional knowledge: the cost when the key engineer leaves	Lock-in: the cost of migrating away in three years

Two practical notes. Use the same horizon for both — three years is usually right for hospital systems. And put a number on the exit: how much would it cost, in money and in months, to stop using this in year three and go back to how you work now? If nobody can answer that, you are not comparing options, you are comparing brochures.

RED FLAGS

- The comparison includes licence cost but not integration cost.
- The build estimate has no maintenance line.
- “We will build it ourselves” is proposed by people who will not maintain it.
- The skip option was never seriously discussed.
- The exit cost has never been estimated.

CHAPTER 5 TAKEAWAY

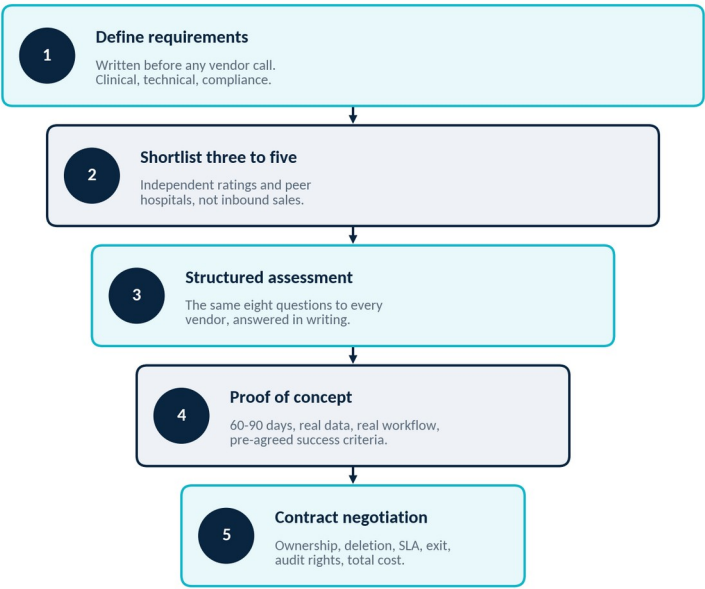
Three answers, not two. Build for genuine differentiation, buy for parity, and skip without embarrassment when the need, the data or the return does not hold up. Compare total cost of ownership over the same horizon on both sides, and always price the exit.

Chapter 6 — Vendor Selection and Procurement

A funnel, not a shootlist

Procurement done badly is a series of demonstrations followed by a decision. Procurement done well is a funnel that narrows on evidence, where each stage is an opportunity to stop and where the criteria were written before anyone saw a product.

The order matters more than the rigour. Requirements written after the first demonstration will describe that product, and every subsequent evaluation will quietly be scored against a competitor’s feature list.



Every stage is a chance to stop. Stopping early is cheap; stopping late is not.

Figure 6.1 — The five-stage procurement funnel. Every stage is a chance to stop; stopping early is cheap and stopping after signature is not.

Step 1 — Define requirements before you speak to anyone

Write the requirements first, in your own language, from the clinical problem. Cover the clinical requirement — what the tool must do, for whom, at what point in the workflow, and to what standard. Cover the technical requirement — which systems it must integrate with, by what mechanism, and where the data may reside. Cover the compliance requirement — the agreements, assessments and evidence your

organisation will need before deployment. And cover the operational requirement — support hours, response times, and who does what when it breaks at three in the morning.

Mark each requirement as mandatory or desirable, and get the clinical lead to sign the mandatory list. That signature is what protects you later from a product that is impressive at everything except the thing you needed.

Step 2 — Shortlist three to five

Build the shortlist from evidence rather than from inbound attention. Independent healthcare technology rating services, peer hospitals running the same use case, published clinical evaluations and professional networks are all better sources than the vendors who found you first. Speak to hospitals of a similar size and case mix; a reference from a very different institution tells you the product exists but little about whether it will work for you.

Three to five is the right number. Fewer and you have no comparison; more and the assessment becomes superficial for all of them.

Step 3 — Structured assessment

Ask every shortlisted vendor the same questions, in writing, and keep the answers. Written answers are slower to obtain and vastly more useful than a conversation, because they are specific, comparable, and referenceable during contract negotiation.

Table 6.1 — The structured assessment. Ask every vendor the same eight areas and require written answers.

Area	What to ask
Product	Does it meet the mandatory clinical requirements? How is it deployed, and how are updates released and validated?
Data	Where is data stored and processed? Who can access it? Is customer data used to train shared models, and can that be switched off?
Integration	Which of our systems does it integrate with, by what interface, and who has done it before?
Compliance	Which agreements, assessments and independent reports can you provide before signature?
Evidence	What clinical evaluation exists? On which population? Published, peer-reviewed, or internal?
Support	What is included, what are the response commitments, and what is chargeable?
Financial	What is the full three-year cost, including integration and growth in usage? Is the company financially stable?

Area	What to ask
References	May we speak to two current customers of similar size, including one that has had a serious problem?

The last row is the most revealing question in the table. A vendor who can put you in touch with a customer who had a serious problem and stayed is a vendor who handles problems well. A vendor who cannot is telling you something either way.

Step 4 — Proof of concept

Before committing to full deployment, run a proof of concept. Keep it deliberately narrow: a single department or use case, sixty to ninety days, success criteria agreed and written down before it starts, real hospital data under proper governance rather than a vendor-supplied sample, and a test against actual clinical workflow rather than a demonstration environment.

A proof of concept answers a technical question — can this work here? It is not the same thing as the pilot in Chapter 7, which answers a clinical and operational question: does this improve outcomes in our hospital, with our data, in our workflows? Running the first and reporting it as the second is one of the most common and most damaging shortcuts in healthcare AI.

Step 5 — Contract negotiation

Negotiate the terms that matter after the trial, when you have evidence and the vendor has invested effort, and get your compliance and legal colleagues into the drafting early rather than at signature.

Table 6.2 — Contract terms worth the argument. Settle these before signature; none of them improve afterwards.

Term	What to settle in writing
Data ownership	Who owns the input data, the outputs, and any model improved using your data
Retention and deletion	What is retained, for how long, and what is deleted when the contract ends — with evidence of deletion
Training use	Whether your data may be used to improve shared models, and how that is switched off
Service levels	Availability, response and resolution commitments, and the remedy when they are missed
Exit	Notice periods, data return format, transition assistance, and cost
Pricing	Total three-year cost, what triggers an increase, and what happens as usage grows

Term	What to settle in writing
Compliance	Which agreements and independent reports are provided and maintained for the life of the contract
Audit rights	What you may inspect, how often, and what the vendor must disclose following an incident

Procurement checklist

- ☐ Requirements were written before the first vendor conversation
- ☐ Mandatory requirements are signed off by the clinical lead
- ☐ Three to five vendors were assessed against identical questions
- ☐ Answers were provided in writing and retained
- ☐ At least two customer references were spoken to directly
- ☐ A proof of concept ran on real data in a real workflow
- ☐ Compliance reviewed the contract before signature, not after
- ☐ Exit terms, deletion evidence and total three-year cost are all written into the agreement

RED FLAGS

- The requirements document quotes a vendor's feature names.
- Only one vendor was seriously considered.
- The proof of concept ran on vendor-supplied data.
- Compliance saw the contract the week it was signed.
- Nobody has read the clause about training on customer data.
- The reference customers were all chosen by the vendor and all delighted.

CHAPTER 6 TAKEAWAY

Write requirements before you take a demonstration. Ask every vendor the same eight areas in writing. Prove the technology on your own data in your own workflow. Then negotiate ownership, deletion, training use, service levels, exit and audit rights — because after signature you are asking for favours rather than agreeing terms.

PART III

The Pilot

Chapter 7 — The Pilot Playbook

What a pilot is for

A pilot is not a demonstration and it is not a proof of concept. It is a controlled experiment designed to answer exactly one question: does this solution improve outcomes in our hospital, with our data, in our workflows?

Everything about the design follows from taking that sentence literally. An experiment has a hypothesis, a baseline, a defined population, a fixed duration and a pre-agreed way of being wrong. A pilot that cannot fail is not an experiment; it is an installation with a launch date, and it will produce a result that convinces nobody who was not already convinced.

A pilot is not a demo. It is a controlled experiment with a pre-agreed way to fail.

60 - 90 days · one department · 5 - 10 clinicians · real hospital data

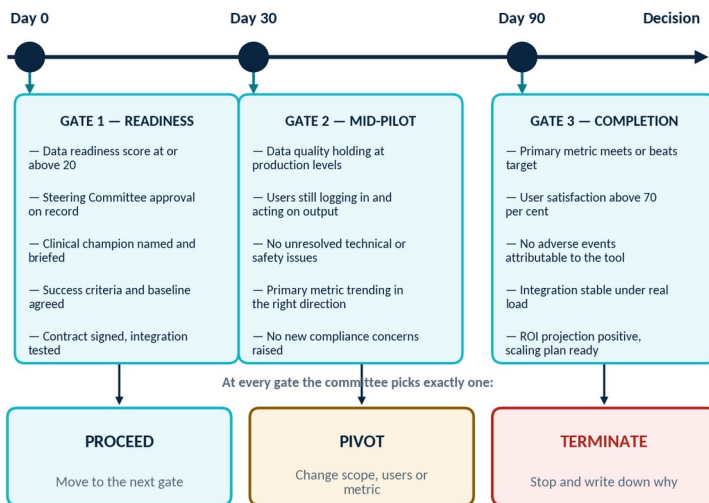


Figure 7.1 — The pilot and its three gates. At each gate the steering committee chooses exactly one of proceed, pivot or terminate — and terminate is a legitimate outcome, not a failure of nerve.

Scope

Constrain the pilot deliberately. One department only — resist every request to include a second, because a pilot spread across departments becomes a deployment with a smaller budget. Sixty to ninety days: long enough for genuine results and for the novelty to wear off, short enough to

hold attention and keep the end date credible. Five to ten clinicians, enough for meaningful patterns without overwhelming the support you can offer. And real hospital data throughout, never synthetic and never vendor-supplied.

The novelty point deserves emphasis. Adoption in the first two weeks of anything is meaningless. What matters is what people are still doing in week eight, when the tool is no longer interesting and the department is busy.

Success criteria, agreed in advance

Before the pilot starts, write down what success looks like: a primary metric with a target and a measured baseline, two or three secondary metrics, the safety measures that must not deteriorate, and the level below which the pilot is judged unsuccessful. Have the steering committee approve it. Circulate it to the participating clinicians.

Table 7.1 — An illustrative success criteria sheet. The structure is the point; the specific metrics belong to your use case.

Criterion	Baseline	Target
Primary — time from imaging study to reported result	Measured over the eight weeks before the pilot	Twenty per cent reduction
Secondary — proportion of urgent findings flagged within one hour	Current departmental figure	Improvement, no deterioration
Adoption — proportion of eligible cases where output was viewed	Not applicable	Above seventy per cent by week six
Satisfaction — participating clinician survey	Not applicable	Above seventy per cent would continue
Safety — adverse events attributable to the tool	Zero	Zero, with any event triggering immediate review

Two disciplines make this work. Measure the baseline before the pilot begins, because a baseline reconstructed afterwards is an argument rather than a measurement. And name the failure threshold explicitly — the level at which you would stop. Committees that have not agreed a stopping point in advance almost never find one afterwards.

The three gates

The pilot is punctuated by three formal decision points. Each is a scheduled steering committee item with a written pack, not a conversation in a corridor.

Gate 1 — readiness, before the pilot starts. The data readiness score is above the threshold. The steering committee has approved the pilot design.

A clinical champion is named and briefed. Success criteria are agreed and the baseline is measured. The contract is signed and the integration has been tested end to end with real users.

Gate 2 — mid-pilot, around day thirty. Data quality is holding at production levels. Users are still logging in and acting on what they see. There are no unresolved technical or safety issues. The primary metric is trending in the right direction. No new compliance concern has surfaced.

Gate 3 — completion, around day ninety. The primary metric meets or beats its target. User satisfaction is above the agreed level. No adverse events are attributable to the tool. The integration has been stable under real load. The return projection is positive on total cost of ownership, and a scaling plan exists.

The gate decision

At each gate the steering committee chooses exactly one of three outcomes, and records the reasoning.

Proceed means continue as designed to the next gate. Pivot means change something specific — the scope, the users, the workflow placement, the metric — and continue with the change written down and the clock reset appropriately. A pivot that is not documented is indistinguishable from drift, and drift is how pilots run for two years.

Terminate means stop, and it deserves rehabilitation as a word. A pilot that terminates at gate two has cost you sixty days and taught you something specific about your data, your workflow or your vendor. A pilot that limps through to gate three because nobody wanted to be the person who stopped it has cost you a year, the goodwill of ten clinicians, and the credibility of the next project.

Gate 1 readiness checklist

- ☐ Data readiness score is above the published threshold
- ☐ Steering committee approval is recorded in the minutes
- ☐ A clinical champion is named, briefed and willing
- ☐ Primary and secondary metrics are defined with targets
- ☐ The baseline has been measured, not estimated
- ☐ The failure threshold is written down and approved
- ☐ The contract is signed and compliance requirements are met
- ☐ Integration has been tested end to end by an actual user
- ☐ The rollback plan exists and names who executes it

RED FLAGS

- The baseline is being reconstructed after the pilot started.
- The pilot has grown to a second department “because they asked”.
- There is no agreed level at which the pilot would be stopped.
- Gate reviews keep being postponed for scheduling reasons.
- Adoption is reported only for the first fortnight.
- The vendor is presenting the results rather than your own analyst.

CHAPTER 7 TAKEAWAY

One department, sixty to ninety days, five to ten clinicians, real data, and three gates with a written decision at each. Measure the baseline first and agree the failure threshold in advance. A pilot you cannot stop is not an experiment, and it will not produce evidence anyone trusts.

Chapter 8 — Integration: Making AI Fit Real Workflows

The tool that nobody opens

The most common cause of low adoption is not scepticism. It is a second login.

Clinical work is dense, interrupted and time-boxed. A tool that lives in its own application, behind its own password, in a separate browser tab, will be used during the training session and rarely afterwards — regardless of how good it is. Integration is not the final technical task of an AI project. It is the project.

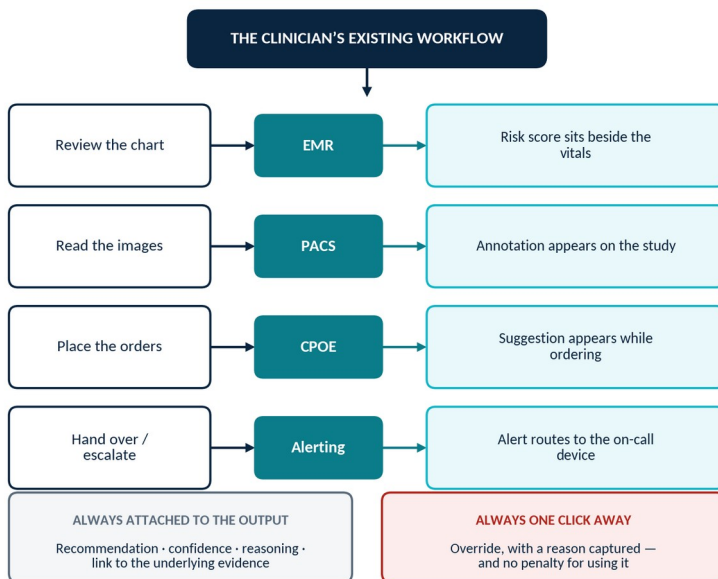


Figure 8.1 — Where AI output has to appear. The rule is simple: the output belongs inside the system the clinician was already using for that decision.

Principle 1 — Meet clinicians where they are

Do not ask clinical staff to switch systems. Put the output into the tools they already have open: risk scores and predictions in the electronic medical record beside the data they relate to; imaging annotations on the study itself in the imaging system; ordering suggestions inside the order entry screen at the moment of ordering; and urgent alerts through the communication system the on-call team already carries.

This is more work for you and less work for them. That trade is always worth making, because your project has one implementation and their day has forty interruptions.

Principle 2 — Minimise clicks

Every additional click reduces adoption, and the effect is much larger than anyone estimates in a design meeting. The target is zero additional clicks: the output appears where the clinician was already looking, at the moment it is relevant, without being asked for.

Where zero is impossible, count the clicks explicitly and treat the number as a design constraint with an owner. “Three clicks” is a requirement you can negotiate down. “It is in the system” is not.

Principle 3 — Always provide context

An output without context is not usable clinically, whatever its accuracy. Every AI output should carry four things: the recommendation itself; a confidence indication expressed in language clinicians actually use; the reasoning, where the model can offer one; and a route to the underlying evidence, whether that is the contributing data, the guideline or the study.

Context is what allows a clinician to disagree intelligently, and the ability to disagree intelligently is what produces trust. A tool that says only “high risk” asks to be either obeyed or ignored, and busy people choose ignored.

Principle 4 — Always allow override

Clinicians must be able to override any AI recommendation easily, with a reason captured, and without penalty or friction. The tool is an input to a professional judgement, not a mandate.

Overrides are also your best monitoring signal. A cluster of overrides on a particular patient type tells you something the accuracy metrics will not, and it usually tells you months earlier. Instrument the override reason field and review it every week during the pilot.

Integration checklist

- ☐ Output appears inside the existing clinical system, not a separate application
- ☐ Output is visible at the point of care, at the moment of the decision
- ☐ Viewing the output requires no additional clicks
- ☐ Confidence and reasoning are shown alongside the recommendation
- ☐ A route to the underlying evidence is one click away
- ☐ The override mechanism is simple, documented and carries no penalty
- ☐ Override reasons are captured and reviewed weekly

- ☐ The output does not interrupt or reorder an existing workflow step
- ☐ Output flows into clinical documentation without re-typing
- ☐ Mobile access works where the workflow is genuinely mobile

RED FLAGS

- Users need a separate password.
- The training material explains how to find the output.
- Confidence is shown as a raw number with no interpretation.
- The override requires a free-text justification longer than the decision.
- Alerts fire for cases the clinician had already identified.
- Nobody has counted the clicks.

CHAPTER 8 TAKEAWAY

Integration is the project, not the last task in it. Put the output where clinicians already look, at zero additional clicks, with confidence, reasoning and evidence attached, and make overriding it easy — then read the overrides, because they are your earliest warning system.

Chapter 9 — Change Management

Twenty per cent technology

AI implementation is roughly twenty per cent technology and eighty per cent people. The proportions are arguable; the direction is not. The best model in the world, integrated perfectly, is worthless if the clinicians it was built for do not use it.

It helps to treat resistance as information rather than as an obstacle. Clinical staff have watched a great many systems arrive promising to save them time, and a fair number of those systems added work instead. Scepticism in that context is not irrational — it is an accurate prior belief, and it will be updated by evidence rather than by enthusiasm.

AI implementation is 20 per cent technology and 80 per cent people

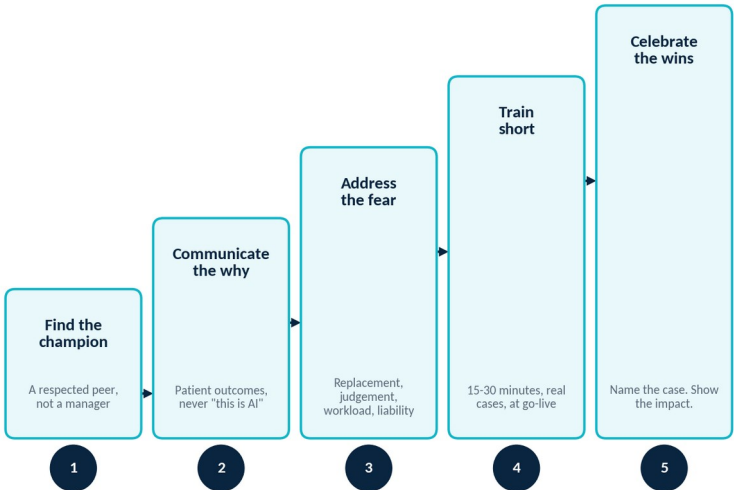


Figure 9.1 — The five steps of clinical change management. They are sequential: skipping the champion and starting at training is the most common and most expensive shortcut.

Step 1 — Find the clinical champion

Every AI project needs a clinical champion: a physician or nurse leader who believes the project is worth doing, has genuine credibility with peers, can explain the value in clinical language, is willing to hear concerns without becoming defensive, and will give you honest feedback rather than reassurance.

The champion is not a project manager and should not be given project management tasks. They are a peer advocate, and their value comes entirely from being seen as a clinician rather than as part of the implementation. Protect that. Also protect their time formally — an unfunded champion is a volunteer, and volunteers get pulled back into clinical work the moment the department is short.

Step 2 — Communicate the why

Clinicians do not care about artificial intelligence. They care about patients, and about getting through a difficult list without missing something. Frame the project in those terms: this helps us identify deterioration earlier; this reduces medication errors; this returns twenty minutes a shift to patient contact.

Avoid the three framings that reliably fail. “This is AI” describes the technology rather than the benefit. “This will improve efficiency” is heard, accurately, as a plan to do more with less. “This is what leadership wants” converts a clinical proposition into a compliance exercise, and compliance exercises get the minimum.

Step 3 — Address the fear directly

Four fears come up repeatedly, and all four are reasonable. Name them out loud before somebody else does; unstated fears do not dissipate, they organise.

Table 9.1 — The four recurring fears. Answer them before deployment, in writing, with the clinical lead present.

The fear	The honest response
“This is a step towards replacing me”	The tool produces an input; the clinical decision and the accountability remain with you. Say this only if it is true, and design so that it is.
“I will lose professional judgement”	Override is one click, carries no penalty, and the override reasons are reviewed by the clinical lead — not by the vendor.
“This will add to my workload”	Show the measured click count and the time study. If it adds work, fix the workflow before deployment rather than defending it.
“Who is liable when it is wrong?”	Set out in advance, in writing, how the tool is intended to be used and what your organisation’s position is. Involve your medico-legal advisers early; do not improvise an answer in a training session.

Step 4 — Train short and late

Training should be short — fifteen to thirty minutes, not a half day. It should be hands-on and built around real cases from the department rather than demonstration data. It should be delivered just before go-live rather

than months in advance, because training delivered early is training delivered twice. And it should continue in small doses: a short refresher after a fortnight, and a note whenever behaviour changes after an update.

Step 5 — Celebrate the specific

When the tool contributes to a good outcome, say so publicly and concretely. Not “adoption is up eleven per cent” but “on Tuesday the flag came up ninety minutes before the on-call team would have reviewed those results, and the patient went to theatre earlier”. Anonymised, accurate, and specific enough to be believable.

One credible story from a respected colleague moves adoption further than any dashboard. This is not manipulation; it is how clinical practice has always propagated.

Reading resistance

Most objections are precise and worth taking literally. Each of the five below is a different problem with a different fix, and treating them all as “change resistance” guarantees you fix none of them.

Table 9.2 — Objections are information. Each of these has a different cause and a different remedy.

What you hear	What it usually means	What to do
“I don’t trust it”	No evidence has been shown on cases they recognise	Show performance on their own department’s cases, including the misses
“This takes too long”	The workflow is genuinely slower	Count the clicks, redesign the placement, re-measure
“It is not accurate enough”	The comparison is against perfection, not against current practice	Show the accuracy data beside the current baseline
“I don’t need help with this”	The framing sounded like a competence judgement	Reframe as a second reader, and let them see the cases it catches
“This is just a fad”	They have seen initiatives abandoned before	Show the governance, the funding commitment and the measurement plan

RED FLAGS

- The champion was appointed rather than volunteered.
- The champion's time is unfunded.
- The first communication to staff described the technology.
- Training happened six weeks before go-live.
- Nobody has answered the liability question in writing.
- Objections are being logged as resistance rather than as defects.

CHAPTER 9 TAKEAWAY

Find a credible peer, fund their time, and let them lead. Frame everything in patient terms. Name the four fears before somebody else does. Train briefly and late. Celebrate specific, verifiable cases. And read objections as design feedback, because that is almost always what they are.

PART IV

The Scale

Chapter 10 — Measuring What Matters

The question behind every metric

Every measurement in this chapter exists to answer one question: did it actually help? Not was it interesting, not was it well received at the board presentation, not is the model accurate in isolation. Did the hospital work better, for patients, because this thing exists.

That question is harder than it looks, because the honest answer requires a baseline you measured beforehand and a willingness to publish a negative result. Both are matters of governance rather than analytics.

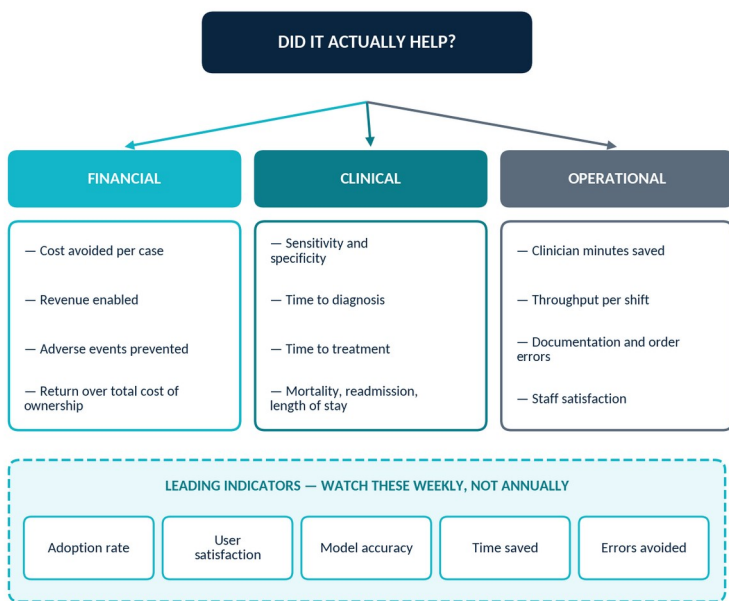


Figure 10.1 — The three families of measurement, and the leading indicators that tell you months earlier what the annual figures will eventually say.

Financial measures

Table 10.1 — Financial measures. State the horizon and the assumptions beside the number; a return figure without them is not reviewable.

Metric	How to calculate it
Cost avoided	Reduction in cost per case, multiplied by the number of cases affected
Revenue enabled	Additional throughput or capacity, valued at the contribution per case

Metric	How to calculate it
Adverse events prevented	Reduction in event rate, multiplied by the average cost per event
Staff time released	Minutes saved per case, multiplied by volume and by the loaded hourly cost
Total cost of ownership	Licensing, integration, training, support and internal effort over three years
Return	$(\text{Benefit} - \text{total cost}) \div \text{total cost}$, stated over the same three-year horizon

Two cautions. Time released is only a saving if the released time is actually redeployed — otherwise it is a benefit to working conditions, which is worth having but should not be counted as cash. And prevented events are estimated by definition; present them as a range with the assumption stated, not as a single confident figure.

Clinical measures

Table 10.2 — Clinical measures. Predictive value depends on prevalence, so a model that performed well elsewhere may perform differently here with identical sensitivity.

Metric	What it tells you
Sensitivity and specificity	How the model performs on your population, not the vendor's
Positive and negative predictive value	What a result actually means at your prevalence
Time to diagnosis	Whether the pathway is genuinely faster
Time to treatment	Whether the faster diagnosis changed anything downstream
Length of stay, readmission, mortality	Whether the outcome moved — the measures that ultimately count

Operational measures

Table 10.3 — Operational measures. The last row predicts sustained adoption better than any other single number.

Metric	What it tells you
Clinician minutes per case	Whether the tool adds work or removes it
Throughput per shift	Whether capacity changed in practice
Documentation and order errors	Whether quality improved where the tool touches the workflow
Workflow steps eliminated	Whether the process was simplified or merely supplemented
Staff satisfaction	Whether the people using it would keep it

Leading indicators

Outcome measures arrive late. Leading indicators arrive weekly, and they are what allow you to intervene while intervention is still cheap. Watch five: adoption rate, defined as the proportion of eligible cases where the output was viewed; user satisfaction, sampled briefly and often rather than surveyed exhaustively once; model accuracy against a rolling recent sample; time saved per case; and errors avoided.

Adoption is the earliest and most reliable warning. It falls before satisfaction scores drop, before complaints reach the steering committee, and long before the outcome measures show anything. A declining adoption curve in week five is a design problem you can still fix in week six.

Reporting honestly

Report against the baseline you measured before the pilot began, state the assumptions behind every projected figure, show the confidence interval or the range rather than a single number, and include the measures that did not improve. A report containing only good news is not evidence — it is advocacy, and experienced boards read it that way.

Have the analysis presented by your own analyst rather than by the vendor. This is not a matter of trust; it is a matter of who is accountable for the number twelve months from now.

RED FLAGS

- The baseline was reconstructed after the fact.
- Return is quoted without a stated time horizon.
- Time saved is counted as cash but nobody redeployed the time.
- Only improving metrics appear in the report.
- Accuracy is quoted from the vendor's published study rather than measured locally.
- Adoption is not being tracked weekly.

CHAPTER 10 TAKEAWAY

Measure financial, clinical and operational effects against a baseline you captured first, over a stated horizon, with assumptions visible. Watch the five leading indicators weekly — adoption above all — because they let you fix problems while fixing them is still inexpensive.

Chapter 11 — Scaling from Pilot to Production

Earning the right to scale

A successful pilot is permission to consider scaling. It is not permission to scale. The gap between the two is where a great deal of money is lost, because the conditions that made a pilot succeed — attentive support, motivated volunteers, a curated dataset, a single friendly department — are precisely the conditions that do not survive expansion.

So before expanding, test yourself against a written list. All ten of the criteria below should be true. Not most of them.

The ten readiness criteria

1. The pilot met its pre-agreed success criteria, measured against the baseline captured beforehand.
2. Clinical validation is complete, on your population, and signed off by the clinical lead.
3. The integration ran stably under real load for the full pilot period.
4. Adoption reached and held above the agreed level after the novelty had worn off.
5. The return projection is positive on total cost of ownership over three years.
6. Compliance requirements are met, documented and re-checked for the wider scope.
7. A support model exists, with named people and defined response times.
8. Monitoring is in place for accuracy, drift, availability and override rates.
9. A rollback plan exists, has an owner, and has been tested rather than written.
10. The budget for the next twelve months is approved, including maintenance.

The two criteria most often waved through are the eighth and the ninth. Monitoring and rollback feel like paperwork during a period of optimism, and they are the only two things that matter on the day something goes wrong at scale.

Scope widens only after the previous phase has proved itself

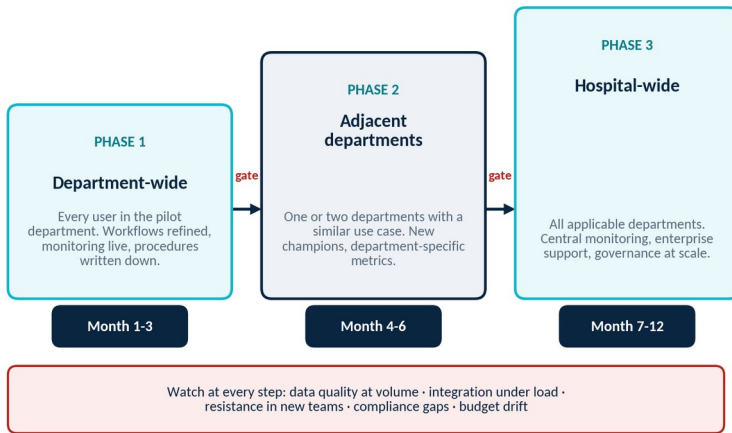


Figure 11.1 — Three phases over twelve months, with a gate between each. Scope widens only after the previous phase has proved itself under real conditions.

Phase 1 — Department-wide, months one to three

Expand to every user in the pilot department rather than the volunteers. This is a bigger step than it appears: the volunteers chose to be there and the rest of the department did not. Expect adoption to be lower and objections to be sharper, and treat both as information.

Use this phase to refine workflows, establish support procedures, bring monitoring fully live, and write down the operating procedures that currently live in one person's head.

Phase 2 — Adjacent departments, months four to six

Expand to one or two departments with a genuinely similar use case. Each new department needs its own champion — a champion does not transfer — and its own baseline, because its starting position will differ. Adapt the workflow to how that department actually works rather than asking it to adopt the first department's habits.

Phase 3 — Hospital-wide, months seven to twelve

Expand to all applicable departments, move to central monitoring and enterprise support, and place the capability formally under the governance structure from Chapter 2 as a standing item rather than a project. At this point it stops being an initiative and becomes part of how the hospital works, which is the intended destination.

The risks that appear only at scale

Table 11.1 — Risks that a pilot cannot surface, and what to do about each before the phase begins.

Risk	How it shows up	Mitigation
Data quality degrades at volume	Accuracy falls as data from new sources arrives	Automated quality monitoring with alert thresholds set before expansion
Integration breaks under load	Latency, timeouts, missed messages at peak hours	Load test at projected peak plus a margin, before the phase begins
Resistance in new departments	Low adoption despite a successful pilot elsewhere	A named local champion and a locally measured baseline for each new area
Compliance gaps at wider scope	New data types, new users, new locations fall outside the original assessment	Re-run the compliance review for every phase, not once at the start
Budget drift	Licensing, support and integration costs rise faster than usage	Model the three-year cost at full scale before phase one, not after phase two

RED FLAGS

- Scaling was approved with three of the ten criteria outstanding.
- The rollback plan has never been tested.
- The new department has no champion of its own.
- Compliance was reviewed once, at the start.
- The support model is still “the project team will handle it”.
- Cost at full scale has never been modelled.

CHAPTER 11 TAKEAWAY

A successful pilot earns you a conversation about scaling, not the scaling itself. Require all ten readiness criteria, expand in three gated phases, give every new department its own champion and its own baseline, and re-run compliance at each phase — because scope, not technology, is what changed.

Chapter 12 — Compliance and Ethics

Four gates in series

This chapter describes four gates that a healthcare AI deployment should pass before it touches patient care. They are in series: a project stops at the first one it cannot pass, and no amount of strength at gate three compensates for a failure at gate one.

A necessary caution before the detail. Regulatory requirements differ by country, by state or region, by the type of organisation you are, and by what the system actually does — and they change. Nothing in this chapter is legal advice, and no framework in a book can substitute for your own compliance officer, your own legal advisers, and the current text of the regulations that apply to you. Use these gates as a structure for the conversation, then verify every specific against the primary source and against the people accountable for it in your organisation.

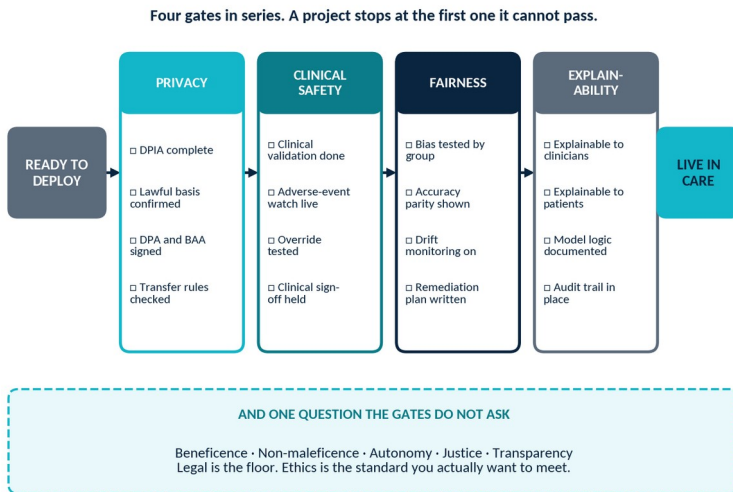


Figure 12.1 — The four compliance gates, and the ethical question that sits underneath all four. Legal compliance is the floor, not the standard.

Gate 1 — Privacy

Establish, in writing and before deployment, which data protection regime or regimes apply to this system and this data. Complete whatever impact assessment your jurisdiction requires. Confirm the lawful basis for the processing, and confirm it separately for any secondary use such as model

improvement. Put the necessary agreements in place with every party that will process the data on your behalf. Check whether data will cross a border, physically or through a support arrangement, and what that requires.

Two questions catch most organisations out. Where exactly is the processing happening, including during support and troubleshooting? And can the vendor use your data to improve models used by other customers — and if the contract permits it, has anyone switched it off?

Gate 2 — Clinical safety

Determine whether the system meets the definition of a medical device under the rules that apply where you operate. This depends on what the software claims to do and how it is used, and it is a question for your regulatory adviser rather than for the vendor's marketing material. If it is a regulated device, obtain and check the evidence of its regulatory status; if it is not, record the reasoning for why not.

Independently of that determination, validate clinical performance on your own population, put adverse-event monitoring and reporting in place before go-live, verify that clinician override works and is used, and obtain formal clinical sign-off from someone who is accountable for it.

Gate 3 — Fairness

Models learn from historical data, and historical healthcare data records historical inequities alongside historical clinical facts. A model that is accurate on average can be substantially less accurate for a subgroup, and average accuracy will not reveal that.

So test performance separately across the demographic groups your population contains, using group definitions agreed with your clinical and governance colleagues. Document any material difference in accuracy. Where a gap exists, write a remediation plan — more representative data, threshold adjustment, restricted scope of use, or a decision not to deploy for that group — and monitor for drift continuously, because a model that is fair at launch may not stay that way as the population changes.

Gate 4 — Explainability

A clinician must be able to understand why a recommendation was made, well enough to agree or disagree with it professionally. A patient, if they ask, must be able to be told in plain language that AI was involved in their care and what part it played. And the organisation must retain a documented account of the model's logic, its version history and an audit trail of its outputs.

Explainability is frequently traded away for accuracy during selection. That trade is sometimes correct, but it should be made deliberately, recorded, and approved by the steering committee — not discovered in month nine when a clinician asks a reasonable question and nobody can answer it.

Beyond compliance

Legal compliance is the floor. Five principles set the standard most hospitals actually want to meet, and they are worth putting into the steering committee charter explicitly.

Beneficence: the system should genuinely improve patient outcomes, not merely improve a metric. Non-maleficence: it should not introduce new harms, including harms of over-reliance and alert fatigue. Autonomy: patients retain meaningful choice, and clinicians retain meaningful judgement. Justice: benefits and risks are distributed fairly rather than concentrating benefit where data was already plentiful. Transparency: everyone affected knows AI is being used, and how.

When the legal question and the ethical question give different answers, the ethical answer is the one your patients will hold you to, and the one you will want on record.

Pre-deployment compliance checklist

- ☐ The applicable data protection regime is identified in writing
- ☐ The required impact assessment is complete and signed
- ☐ Lawful basis is confirmed, including for any secondary use
- ☐ Processor agreements are executed with every party involved
- ☐ Cross-border processing, including support access, is documented
- ☐ The medical device question has been answered by a qualified adviser
- ☐ Clinical validation on the local population is complete
- ☐ Adverse-event monitoring and reporting are live before go-live
- ☐ Subgroup accuracy has been tested and documented
- ☐ A remediation plan exists for any material accuracy gap
- ☐ Drift monitoring is running with defined thresholds
- ☐ Clinician-facing and patient-facing explanations both exist
- ☐ Model logic, version history and audit trail are retained
- ☐ The five ethical principles are recorded in the committee charter

RED FLAGS

- The compliance review happened after the contract was signed.
- Nobody can say where the data is processed during vendor support.
- Subgroup performance has never been tested.
- The medical device question was answered by the vendor.
- Explainability was traded away without a recorded decision.
- Compliance is treated as a one-time gate rather than an ongoing obligation.

CHAPTER 12 TAKEAWAY

Four gates in series: privacy, clinical safety, fairness and explainability. Verify every specific requirement against the primary source and your own compliance officer, because the rules differ by jurisdiction and they change. Then hold yourself to the five ethical principles, which are the standard your patients will actually judge you against.

PART V

When Things Go Wrong

Chapter 13 — Handling Failure

Failure is data

Not every AI project will succeed. That is not a defect of your programme; it is what running experiments means. The distinguishing feature of a mature organisation is not that it avoids failure but that it recognises failure early, responds proportionately, and extracts something reusable from it.

The alternative is the pattern described in Chapter 1: projects that nobody stops, that fade rather than end, and that leave behind no record of what was learned — so the next team repeats the same mistake with a different vendor and a larger budget.

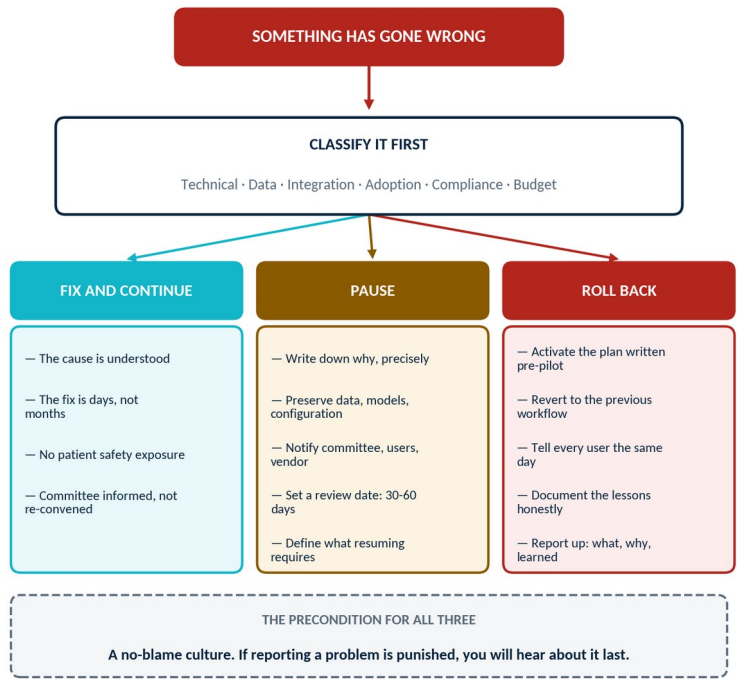


Figure 13.1 — Classify first, then choose one of three responses. All three depend on a culture in which reporting a problem is safe.

Classify before you respond

The response depends entirely on the type of failure, so classify it before deciding anything. The six categories below cover most of what actually happens.

Table 13.1 — Failure taxonomy. Compliance failures are the only category with no “continue while we investigate” option.

Type	What it looks like	First response
Technical	Accuracy below target, unacceptable latency, instability	Retrain, retune or replace; if the cause is not understood within days, escalate
Data	Quality degrades at volume, or the pipeline breaks silently	Fix at source, re-score readiness, add automated quality monitoring
Integration	The workflow does not fit; the output arrives too late to act on	Redesign the placement with the clinical champion, then re-measure
Adoption	Clinicians will not use it despite it working	Return to Chapter 9; treat objections as design defects, not resistance
Compliance	A requirement cannot be met, or scope has outgrown the assessment	Pause immediately; resume only when the gate is genuinely passed
Financial	Costs exceed benefits, or the projection no longer holds	Re-run total cost of ownership honestly; consider terminating

Fix and continue

The lightest response is appropriate when the cause is understood, the fix is measured in days rather than months, there is no patient safety exposure, and the steering committee can be informed rather than reconvened. Record the issue and the fix anyway — a sequence of small fixes is itself a signal worth noticing.

The pause protocol

Pausing is the underused middle option, and it is often the right one. Write down precisely why the pause is happening — vaguely documented pauses become quiet abandonments. Preserve the data, models and configuration so that resuming is possible. Notify the steering committee, the users and the vendor on the same day. Set a review date, typically thirty to sixty days out. And define, in advance, what specifically would have to be true to resume.

That last item is what separates a pause from a slow death. If nobody can state the resumption condition, the project has already ended and the organisation has not admitted it.

The rollback protocol

Rollback is for the cases where the system must come out of the clinical workflow now: a safety concern, a compliance failure, or an instability that cannot be contained. It works only if it was designed before deployment,

which is why it appears as a gate one requirement in Chapter 7 and a readiness criterion in Chapter 11.

Activate the plan written before the pilot. Revert to the previous workflow, which must still be functional — do not decommission the old process until the new one has been stable for a full phase. Tell every affected user the same day, in plain language, including why. Preserve the evidence for analysis. Then document what was learned and report it upward honestly.

The precondition: a no-blame culture

None of the three responses works in an organisation where reporting a problem is punished. If raising a concern about an executive's favoured project is career-limiting, you will hear about problems last, at the worst possible moment, from someone outside the organisation.

Building the alternative is mostly a matter of behaviour rather than policy. Reward early identification of problems visibly. Separate the decision from the person who made it when reviewing what went wrong. Publish lessons from failed projects with the same seriousness as results from successful ones. And make sure the first project that gets terminated is terminated well and publicly, because everyone in the organisation is watching to learn what actually happens.

Post-failure review checklist

- ☐ The failure type has been classified using the taxonomy
- ☐ The root cause is documented, not just the symptom
- ☐ Early warning signs are identified and recorded for next time
- ☐ What would have caught it sooner is written down
- ☐ Costs incurred and value salvaged are both recorded
- ☐ Reusable assets — data work, integrations, procedures — are catalogued
- ☐ The lessons are published where the next project team will find them
- ☐ No individual is named as the cause in the written record

CHAPTER 13 TAKEAWAY

Classify the failure before choosing a response. Fix, pause or roll back — and define the resumption condition whenever you pause. Design the rollback before deployment, keep the old workflow alive until the new one has proved itself, and terminate your first failed project openly, because that is the moment your organisation learns whether honesty is safe.

Chapter 14 — The Twelve-Month Roadmap

One capability, not twelve pilots

This chapter puts the whole book onto a calendar. The roadmap assumes an organisation starting from nothing — no committee, no assessment, no project — and it aims at a single outcome: one proven, governed, measured AI capability in production at the end of twelve months.

That is a deliberately modest target, and it is more than most organisations achieve. A hospital with one working capability and a functioning governance structure is in a far stronger position than one with six pilots and no production deployments, because the second hospital has spent more, learned less, and taught its clinicians that AI projects do not finish.

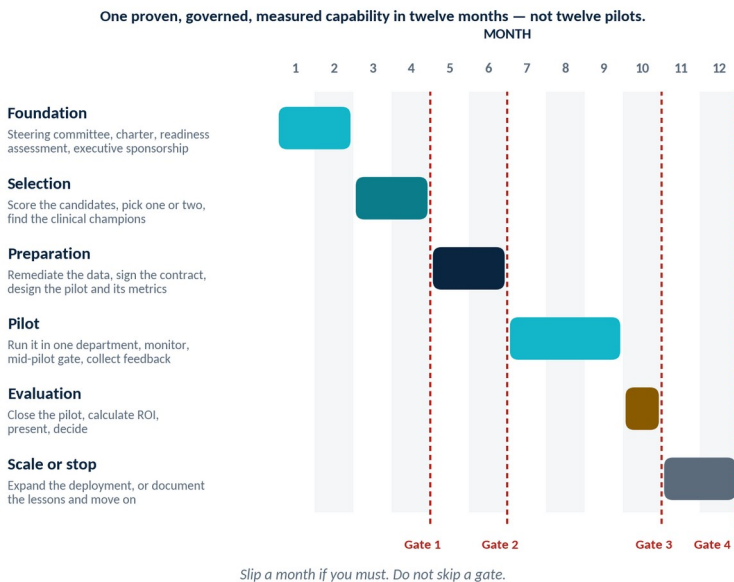


Figure 14.1 — The twelve-month roadmap. Six workstreams, four gates. Slip a month if you must; do not skip a gate.

Months one and two — Foundation

Form the AI steering committee and write its charter. Secure executive sponsorship with a named individual and a budget line rather than general encouragement. Complete an organisational readiness assessment covering data, skills and infrastructure. Establish the governance rule that no AI project proceeds without committee approval, and communicate it.

The deliverable is a charter, a named sponsor, a budget and a recorded rule. If month two ends without those four things, do not move on — everything downstream depends on them.

Months three and four — Selection

Gather candidate projects from across the organisation. Score each against the selection matrix from Chapter 3. Assess data readiness for the leading candidates. Select one or two projects, and — equally important — record why the others were deferred or rejected. Identify clinical champions for the selected projects and confirm they are willing.

Gate 1 falls at the end of month four: the steering committee formally approves the project or projects to proceed.

Months five and six — Preparation

Remediate the data gaps identified during scoring. Run the procurement process from Chapter 6, or begin the build if that was the decision. Design the pilot: scope, participants, duration, success criteria and baseline. Measure the baseline. Complete the compliance pre-work so that gate one of Chapter 7 is a formality rather than a scramble.

Gate 2 falls at the end of month six: pilot readiness confirmed, or the pilot is delayed until it is.

Months seven to nine — Pilot

Run the pilot in one department. Monitor the leading indicators weekly. Hold the mid-pilot gate at day thirty and act on it. Collect structured feedback from participating clinicians rather than impressions. Refine the workflow placement as you learn.

Month ten — Evaluation

Close the pilot on the date you said you would. Analyse the results against the criteria agreed beforehand, calculate the return on total cost of ownership, and present to the steering committee — including the measures that did not improve.

Gate 3 falls here: scale, pivot or terminate, decided on evidence and minuted.

Months eleven and twelve — Scale or stop

If the decision is to scale, begin phase one from Chapter 11: department-wide expansion, with monitoring, support procedures and a tested rollback in place. If the decision is to stop, document the lessons properly, catalogue

the reusable assets, and return to the deferred candidate list with a year of real experience behind you.

Gate 4 falls at the end of month twelve, and it is a genuine decision point rather than a milestone: continue scaling, pause, or conclude.

If you take one thing from this book

Take this: the difference between hospitals that succeed with AI and hospitals that do not is almost never the technology. It is governance that decides, data that is honestly assessed, integration that respects how clinicians actually work, measurement against a baseline captured beforehand, and the institutional courage to stop things that are not working.

Every one of those is a management practice rather than a technical capability. Which is encouraging, because it means the hardest part of healthcare AI is something your organisation already knows how to do — it simply has to choose to do it here as well.

CHAPTER 14 TAKEAWAY

Twelve months, six workstreams, four gates, one capability in production. Foundation before selection, selection before preparation, preparation before the pilot, and evidence before scale. Slip the schedule if you must; never skip a gate.

APPENDICES

Reference Material

Glossary

Terms are defined as they are used in this book. Where a term also has a specific regulatory meaning in your jurisdiction, that meaning governs.

Table A.1 — Glossary of terms used in this book.

Term	Definition as used here
Adoption rate	The proportion of eligible cases in which the AI output was actually viewed or acted upon by a clinician
Baseline	The measured value of a metric before the intervention begins, captured in advance rather than reconstructed
Clinical champion	A respected clinician who advocates for the project among peers and gives the team honest feedback
Data readiness	The assessed suitability of available data for an AI project, scored across availability, quality, accessibility, governance and volume
Drift	Gradual degradation of model performance as the population, the data or the workflow changes over time
Gate	A scheduled decision point at which a governing body chooses explicitly to proceed, change or stop
Integration	Placing AI output inside the clinical systems and workflows that clinicians already use, rather than alongside them
Override	A clinician's documented decision not to follow an AI recommendation, captured with a reason
Pilot	A controlled experiment testing whether a solution improves outcomes in your hospital, with your data, in your workflows
Proof of concept	A narrower technical test of whether a solution can work in your environment at all; not a substitute for a pilot
Total cost of ownership	The full cost of a capability over a stated horizon, including integration, training, support, maintenance and exit

Further Reading

Rather than listing document titles, which change, are superseded and are versioned differently across regions, this section names the kinds of source that are worth your time and where they come from. Search the issuing body's own website for the current version, and confirm applicability with your compliance officer before relying on anything.

- National and regional data protection authorities — for the privacy rules that apply to you, including guidance specific to health data and to automated processing.
- Medicines and medical device regulators in your jurisdiction — for how software is classified, what evidence is required, and how post-market obligations apply to adaptive systems.
- National standards bodies and international standards organisations — for management-system and risk-management standards covering artificial intelligence, information security and medical device software.
- National measurement and technology institutes — for risk-management frameworks that translate well into hospital governance language.
- International and national health bodies — for ethics and governance guidance on artificial intelligence in health.
- Health informatics professional associations — for adoption models, maturity frameworks and practitioner case studies.
- Peer-reviewed clinical informatics and medical journals — for validation studies on the specific use case you are considering, ideally in a population comparable to yours.
- Your own national health service or ministry — for procurement frameworks, clinical safety standards and any mandated assurance process.

One practical note. When a vendor cites a standard, a certification or a regulatory clearance, ask for the document itself, check the scope statement, and check the date. Scope and date are where most of the meaning lives, and neither appears on a slide.

About the Author

Mohammed Imthiyaz A is a senior quality assurance engineer and business analyst working in healthcare information technology. His work sits at the point where clinical software meets the people who have to use it: validating systems before they reach patients, translating clinical requirements into something engineers can build, and testing whether what was built actually does what the workflow needs.

That vantage point is the origin of this book. Most of the failures described here are not exotic; they are ordinary, repeated, and visible well before anyone acknowledges them. The remedies are similarly ordinary — write the criteria down first, measure the baseline, ask the awkward question at the gate rather than after it.

He writes the Practical AI in Healthcare series for the people who have to make these decisions in real organisations, with real constraints and real clinical risk, rather than for a conference audience.

Corrections, disagreements and accounts of what happened when you tried this are all genuinely welcome, and they shape later editions: imthiyazzilaan@gmail.com

A Closing Note on Accuracy

This book describes management practice, not law. Regulatory requirements for healthcare artificial intelligence differ by jurisdiction, by organisation type and by the intended use of the system, and they are being revised continuously. Every specific obligation mentioned here should be verified against the current primary source and confirmed with your own compliance, legal and clinical governance colleagues before you act on it.

Where a statistic would have made a point more forcefully, and the figure could not be verified from a primary source, it has been left out rather than repeated. Frameworks, thresholds and scoring bands in this book are practical starting points intended to be adapted to your organisation — not standards, and not benchmarks.

Nothing here constitutes legal, regulatory, clinical or financial advice.